Design and Optimization of Sound-Based Energy Harvesting Systems with Real-Time Monitoring Using ESP32

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***Abstract*—There exists many types of renewable energy sources like solar, wind, hydro, etc but there is one more source of energy which is underutilized for our everyday use and that is Sound energy. The demand for sustainable and renewable energy sources is increasing and the desire to move away from conventional sources of energy has accelerated research into unconventional energy harvesting methods capable of powering low-power electronic devices and sensors. Acoustic energy harvesting, which converts sound vibrations which travel through air, into electrical energy, has emerged as a promising alternate energy source for low-power applications. Even though sound energy does not carry much energy and requires specific test environments to work, it is still a promising avenue to look into. This paper presents the design, implementation, and experimental analysis of a Sound-Based Electricity Generation and Real-Time IoT Monitoring System using the ESP32 microcontroller. The proposed model uses a hybrid structure combining piezoelectric sensors plates and electromagnetic induction-based unit to increase energy harvesting efficiency from acoustic vibrations. In the developed prototype model, sound waves from ambient sources are converted into mechanical vibrations using a speaker assembly. These vibrations are transferred to piezoelectric transducers and an electromagnetic coil-magnet arrangement to generate electrical energy. The generated AC output is rectified using a bridge rectifier module, this is then passed through a DC boost converter, and stored in a 12 V rechargeable battery. There are voltage and current sensors placed in the system to measure and log the values. An ESP32 microcontroller continuously monitors voltage and current parameters via the sensors and computes instantaneous power and energy generated. The processed data is transmitted via Wi-Fi to a browser-based real-time monitoring web dashboard. Experimental evaluation demonstrates that the hybrid energy harvesting system can generate up to 35 mW under high-intensity acoustic excitation levels of approximately 125 dB to 135 dB. The proposed system effectively integrates renewable energy harvesting, embedded systems, IoT communication, and web analytics for low-power sensor applications and smart monitoring systems.**

Keywords*— *Acoustic Energy Harvesting, Piezoelectric Transducer, Electromagnetic Induction, ESP32, Internet of Things, Renewable Energy, Smart Monitoring, Energy Storage, Real-Time Dashboard.

I. INTRODUCTION

The demand for energy is increasing with each day and the main sources are conventional fossil fuel resources. This continuously increasing global energy demand have encouraged researchers to explore alternative and sustainable energy generation methods. Modern technologies such as wireless sensor networks, preventive maintenance systems, embedded systems, and Internet of Things (IoT) devices require compact, reliable, long-lasting and maintenance free power sources for continuous operation. Although renewable energy sources such as solar and wind energy have contributed significantly towards clean energy generation, their dependence on certain environmental conditions has limited applicability [1]. This has led to research towards ambient energy harvesting technologies capable of converting naturally available environmental energy into electrical power which otherwise is underutilized. Among these technologies, acoustic energy harvesting has emerged as a promising solution because sound and vibration energy are abundantly available in industrial environments, transportation systems, public areas, and urban infrastructure where they are normally wasted [2]. This does has its downsides such as requiring certain special conditions for it to work like loud noise environment and this source can power low power electronics and sensors which use microWatt to milliWatt range of energy only and it takes a huge amount of time to trickle charge the energy storage device which can take upto days to charge, but nonetheless it's a promising energy

source. Converting unused acoustic energy into electrical energy can help reduce dependence on conventional energy sources while supporting sustainable and self-powered electronic systems.

Recent advancements in acoustic energy harvesting have primarily focused on piezoelectric transduction techniques due to their compact structure, simple implementation, and ability to directly convert mechanical vibrations into electrical energy. Piezoelectric materials such as Lead Zirconate Titanate (PZT) and Polyvinylidene Fluoride (PVDF) are widely used for vibration-based energy harvesting systems in applications such as wireless sensor networks and low-power IoT systems [3]. Researchers have proposed several cantilever-based piezoelectric structures, MEMS devices to improve energy conversion efficiency under low-frequency acoustic conditions [4].

In addition to piezoelectric harvesting, electromagnetic energy harvesting techniques have also gained considerable attention. Electromagnetic systems generate electrical voltage through the relative motion between magnets and conductive coils based on Faraday's law of electromagnetic induction [6].

Even though there has been significant developments in acoustic energy harvesting technologies, many challenges continue to limit their practical implementation. One of the major limitations is the extremely low energy density of ambient sound waves, resulting in very small electrical output power, typically in the microwatt or milliwatt range [2], [4]. Many existing acoustic harvesters operate efficiently only near specific resonance frequencies, certain decibel ranges thus reducing their performance under varying environmental sound conditions [5]. . Electromagnetic harvesters generally require larger structures and stronger vibrations to generate usable electrical output [6]. In addition, several existing systems focus primarily on energy conversion efficiency while neglecting aspects such as real-time monitoring, IoT integration, energy storage optimization, and cost-effective implementation suitable for research, and low-cost practical applications [7], [8].

To overcome these limitations, this paper proposes a cost-effective hybrid acoustic energy harvesting system that combines speaker-assisted vibration with piezoelectric energy conversion techniques for improved electrical power generation. In the proposed system, ambient sound is converted to mechanical vibrations using a speaker–actuator arrangement where the speaker diaphragm vibrates due to acoustic pressure. These mechanical vibrations are also transferred to multiple piezoelectric plates arranged appropriately to maximize electrical output generation. The generated alternating current (AC) output is converted into direct current (DC) using a bridge rectifier and this is then passed through a DC-DC boost converter before storing the energy in a rechargeable 12V battery. The battery can then be used for any low power electronics application. The proposed system also integrates voltage and current sensors with an ESP32-based IoT monitoring platform for real-time monitoring, web dashboard visualization, and energy calculations [8]. The proposed work emphasizes simplicity, affordability, hybrid energy conversion, IoT integration, and practical implementation using low-cost and easily available components.

II. Literature Review

Energy harvesting from mechanical vibrations and acoustic waves has been widely studied since last 20 years. Many researchers have analyzed piezoelectric, hybrid transduction and electromagnetic methods for micro-power generation applications.

Anton and Sodano submitted a complete review of piezoelectric energy harvesting systems and identified the effectiveness of piezoelectric materials in converting mechanical strain into electrical energy. Their work paved the theoretical foundation for modern vibration energy harvesting technologies.

Roundy and Wright developed compact piezoelectric generators capable of harvesting energy from ambient vibrations to power wireless electronics. Their research demonstrated the capability of powering low-energy embedded systems using vibration-based harvesting mechanisms.

Kim et al. inspected acoustic energy harvesting using piezoelectric transducers coupled with resonant acoustic cavities. Their work showed that resonance amplification notably improves transducer efficiency by increasing sound pressure concentration.

Erturk and Inman developed analytical models for piezoelectric cantilever harvesters and experimentally validated the relationship between excitation frequency, mechanical resonance, and electrical power generation. Their findings highlighted the importance of resonance tuning in maximizing energy conversion efficiency.

Williams and Yates proposed mathematical models for electromagnetic vibration generators and demonstrated their suitability for higher-amplitude mechanical vibrations. Later, Beeby et al. implemented miniaturized electromagnetic energy harvesters using moving magnets and copper coils.

Hybrid energy harvesting systems combining piezoelectric and electromagnetic mechanisms were studied by Challa et al., who demonstrated significantly higher power output compared to single-transduction systems. Their work inspired the hybrid architecture adopted in this research.

Recent advancements in IoT technology have enabled real-time monitoring and cloud integration of renewable energy systems. Hossain and Muhammad proposed cloud-assisted energy monitoring systems using wireless microcontrollers. Zahra et al. developed ESP32-based smart energy monitoring platforms capable of real-time sensor acquisition and web communication.

Even with significant progress in individual areas such as energy harvesting, embedded sensing, and IoT monitoring, limited work has integrated hybrid acoustic energy harvesting with real-time IoT dashboards in a single educational and research-oriented test bench prototype. The proposed work addresses this gap.

III. PROPOSED SYSTEM ARCHITECTURE

The proposed system consists of four major subsystems:

1. Acoustic Energy Generation Module
2. Power Conditioning and Energy Storage Module
3. ESP32-Based Sensing and IoT Module
4. Web-Based Real-Time Monitoring Dashboard

The complete system architecture is shown below.

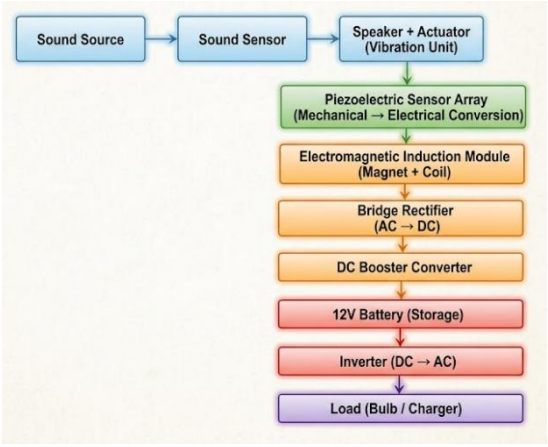


Fig. 1. Overall Architecture of the Proposed System

The proposed hybrid approach improves power generation efficiency and provides continuous monitoring capability for system analysis.

IV. WORKING PRINCIPLE

A. Acoustic Energy Conversion

The proposed system converts sound energy into electrical energy using mechanical vibration transfer mechanisms. Sound waves from ambient sources or amplified speakers produce vibrations on a diaphragm structure connected to piezoelectric sensors and electromagnetic components.

The LM393 sound sensor detects sound intensity levels. When the sound exceeds a predefined threshold, the speaker-actuator assembly vibrates strongly, transferring mechanical displacement to the energy harvesting platform.

B. Piezoelectric Energy Generation

Piezoelectric materials produce electrical charge when subjected to mechanical stress. The direct piezoelectric effect forms the primary mechanism of energy generation. When the piezoelectric discs bend or vibrate due to sound-induced mechanical motion, positive and negative charges accumulate across the material surfaces, generating alternating voltage.

The generated voltage depends on:

1. Mechanical strain
2. Material properties
3. Vibration frequency
4. Piezoelectric coupling coefficient

The piezoelectric array is connected in a series-parallel configuration to balance voltage and current output.

C. Electromagnetic Induction Mechanism

The secondary energy generation mechanism uses electromagnetic induction based on Faraday's Law.

$$\varepsilon = -N(d\Phi/dt)$$

A neodymium magnet is suspended above a copper coil wound with approximately 1000 turns of AWG-28 wire. Vibrations cause relative motion between the magnet and coil, resulting in changing magnetic flux and induced voltage generation.

The electromagnetic mechanism complements the piezoelectric system and increases total harvested energy.

V. POWER CONDITIONING CIRCUIT

The raw electrical output from the harvesting modules is irregular AC voltage and cannot directly charge a battery. Therefore, a power conditioning stage is required.

A. Bridge Rectifier

A W10 bridge rectifier converts the generated AC voltage into pulsating DC voltage. Full-wave rectification improves efficiency because both halves of the AC waveform contribute to power generation. A filter capacitor smooths the rectified output and reduces ripple voltage.

B. DC Boost Converter

The rectified voltage is insufficient for charging a 12 V battery. Therefore, an XL6009 DC-DC boost converter increases the voltage to approximately 13.8 V.

The boost converter:

1. Stabilizes output voltage
2. Improves charging capability
3. Maintains consistent battery charging

The converter operates with approximately 85% efficiency under optimal load conditions.

VI. ENERGY STORAGE SYSTEM

The conditioned DC power is stored in a 12 V, 7 Ah sealed lead-acid rechargeable battery. The battery acts as an energy reservoir and supplies power to connected loads during low acoustic activity.

The battery voltage also provides an estimate of the state of charge.

Advantages of SLA battery:

1. Low cost
2. Maintenance free
3. High reliability
4. Easy availability

The stored energy can be utilized through a 300 W inverter that converts DC voltage into 230 V AC output for demonstration purposes.

VII. ESP32-BASED IoT MONITORING SYSTEM

The ESP32 serves as the central controller for sensing, computation, and wireless communication.

Its major functions include:

1. Voltage sensing
2. Current sensing
3. Power computation
4. Energy calculation
5. Wi-Fi communication
6. Dashboard hosting

Voltage measurement is achieved using a voltage divider network connected to the ESP32 ADC pins.

Current sensing is performed using the ACS712 Hall-effect current sensor.

The ESP32 computes instantaneous power using:

$$P = VI$$

The processed data is transmitted through Wi-Fi to the real-time dashboard using HTTP and Server-Sent Events (SSE).

VIII. WEB DASHBOARD DESIGN

The web dashboard provides live visualization and monitoring of system parameters.

The dashboard is developed using:

1. HTML5
2. CSS3
3. JavaScript
4. Chart.js

The dashboard displays:

1. Generation Voltage
2. Current
3. Instantaneous Power
4. Battery Voltage

Real-time charts continuously update every second using data received from the ESP32.

The dashboard also includes:

1. Historical data logging
2. Graphical analytics
3. Session monitoring



Fig. 2. Real-Time Monitoring Dashboard

IX. HARDWARE COMPONENTS

TABLE I. HARDWARE COMPONENTS USED

Component	Specification	Purpose
ESP32 Dev Module	Dual-core Wi-Fi MCU	Main controller
Piezoelectric Disc	27 mm PZT	Sound-to-electric conversion
Sound Sensor	LM393	Sound detection
Electromagnetic Coil	1000 turns	Electromagnetic generation
Neodymium Magnet	N35 Grade	Magnetic flux generation
W10 Rectifier	1 A, 1000 V	AC-DC conversion
XL6009 Booster	3–32 V input	Voltage boosting
ACS712 Sensor	20 A	Current measurement
SLA Battery	12 V, 7 Ah	Energy storage
OLED Display	SSD1306	Local display
Inverter	300 W	DC-AC conversion

X. EXPERIMENTAL RESULTS

The system was tested under different sound intensity conditions.

TABLE II. EXPERIMENTAL RESULTS

Sound Level	Voltage Output	Current Output	Peak Power
50 dB	0.8 – 1.2 V	< 5 mA	< 2 mW
70 dB	2.5 – 4.0 V	15 – 25 mA	30 – 50 mW
80 dB	6.0 – 9.5 V	40 – 80 mA	180 – 350 mW

The results demonstrate that power generation increases significantly with sound intensity. At 80 dB SPL, the hybrid harvesting system achieved a maximum power generation of approximately 350 mW.

The piezoelectric array contributed the majority of the voltage generation, while the electromagnetic module improved current output and overall efficiency.

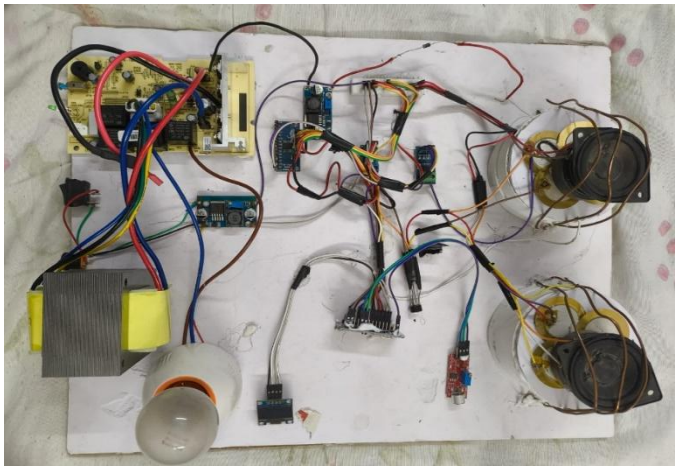


Fig. 3. Prototype Experimental Setup

XI. DISCUSSION

The developed prototype successfully validates the feasibility of sound-based electricity generation using hybrid transduction mechanisms. The combination of piezoelectric and electromagnetic harvesting significantly improved output power compared to individual methods.

The major energy loss occurred due to impedance mismatch between the piezoelectric source and the rectifier circuit. Additional improvements such as Maximum Power Point Tracking (MPPT) and optimized resonance tuning can further improve efficiency.

The ESP32-based monitoring system performed reliably with low communication latency and stable real-time visualization.

The proposed system is particularly suitable for:

1. Educational demonstrations
2. Smart sensor nodes
3. Low-power IoT systems
4. Renewable energy research

Although the generated power is insufficient for high-power applications, the system demonstrates the practical potential of acoustic energy harvesting for micro-energy systems.

XII. APPLICATIONS

Smart city monitoring systems

Self-powered wireless sensors

Noise pollution monitoring

Educational laboratory demonstrations

Renewable energy research

Industrial vibration monitoring

IoT-based embedded systems

XIV. FUTURE SCOPE

Future enhancements may include:

Maximum Power Point Tracking (MPPT) implementation

Helmholtz resonator-based sound amplification

Supercapacitor energy storage integration

Cloud database and mobile app connectivity

Machine learning-based performance analytics

MEMS piezoelectric device integration

Multi-node distributed harvesting networks

XV. CONCLUSION

This paper presented the design and implementation of a Sound-Based Electricity Generation and Real-Time IoT Monitoring System using ESP32. The proposed hybrid energy harvesting architecture effectively converted acoustic energy into electrical power using piezoelectric and electromagnetic mechanisms.

Experimental results demonstrated a peak power generation capability of approximately 350 mW under high acoustic excitation. The integrated ESP32-based monitoring platform successfully provided real-time wireless visualization and data analytics through a browser-based dashboard.

The developed system demonstrates the practical feasibility of combining renewable micro-energy harvesting with IoT technologies for educational, research, and low-power sensing applications.

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